

D667

Precalculus

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

An engineer models a process with $f(x) = 2x + 3$ and $g(x) = x^2 - 1$. Her team uses a spreadsheet with 12 columns, and the data were collected on a Tuesday. What is $f(g(3))$?

- A. 19
- B. 72
- C. 80
- D. 8

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A shipping company converts a package's weight x to a cost with $f(x) = 3x - 5$ dollars. The company has 40 trucks and was founded in 1998. What is $f^{-1}(7)$?

- A. $2/3$
- B. 4
- C. $1/16$
- D. 16

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

For a one-to-one function h , you know that $h^{-1}(10) = 4$. Which statement must be true?

- A. $h(4) = 10$
- B. $h(10) = 10$
- C. $h(4) = 1/10$
- D. $h(10) = 4$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A bacteria culture starts with 500 cells and doubles every 3 hours. The lab keeps it at 37°C in a room with 6 incubators. How many cells are there after 12 hours?

- A. 2,500
- B. 2,000
- C. 8,000
- D. 2,048,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A Ferris wheel's position is described using angles on a unit circle. The ride is 40 meters tall and carries 36 cars. What is $\cos(2\pi/3)$?

- A. $-1/2$
- B. $1/2$
- C. $\sqrt{3}/2$
- D. $-\sqrt{3}/2$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A sound wave in a studio is modeled by $y = 3 \sin(4x) + 1$. The studio has foam panels and was built last year. What are the amplitude and period?

- A. Amplitude 4, period $\pi/2$
- B. Amplitude 4, period 2π
- C. Amplitude 3, period $\pi/2$
- D. Amplitude 3, period 8π

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

Which function has amplitude 5 and period π ?

- A. $y = 5 \sin(x/2)$
- B. $y = 5 \sin(2x)$
- C. $y = 5 \sin(\pi x)$
- D. $y = 2 \sin(5x)$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A student club sells two items. Their records give $2x + 3y = 12$ and $x - y = 1$. The club meets on Thursdays and has 15 members. What is the solution (x, y) ?

- A. (6, 0)
- B. (1, 0)
- C. (2, 3)
- D. (3, 2)

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

For what value of k does the system $2x + y = 4$ and $4x + 2y = k$ have infinitely many solutions?

- A. 4
- B. Any k , since the slopes match.
- C. 2
- D. 8

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A park designer maps a circular fountain with $x^2 + y^2 - 6x + 4y - 12 = 0$, where units are meters. The park opened in May and has 3 entrances. What are the center and radius?

- A. Center $(3, -2)$, radius 25
- B. Center $(-3, 2)$, radius 5
- C. Center $(3, -2)$, radius $\sqrt{12}$
- D. Center $(3, -2)$, radius 5

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A student saves \$5 in week 1 and increases her deposit by \$4 each week. She keeps the money in a blue jar and works at a café on weekends. How much does she deposit in week 20?

- A. \$80
- B. \$81
- C. \$860
- D. \$85

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A geometric sequence starts 3, 6, 12, ... What is the sum of the first 6 terms?

- A. 192
- B. 96
- C. 189
- D. 63

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

An engineer models a process with $f(x) = 2x + 3$ and $g(x) = x^2 - 1$. ~~Her team uses a spreadsheet with 12 columns, and the data were collected on a Tuesday.~~ What is $f(g(3))$?

Target word: evaluate $f(g(3))$ **Decides:** $g(3) = 8$ first, then $f(8)$.

A	Correct	$g(3) = 9 - 1 = 8$; $f(8) = 16 + 3 = 19$.
B	Confusion set	$f(3) \cdot g(3) = 9 \times 8$ multiplies instead of composing.
C	Inverted operation	That is $g(f(3)) = g(9) = 80$; wrong order.
D	Dropped element	Stops at $g(3)$ without applying f .

2. Answer B Concept 2

A shipping company converts a package's weight x to a cost with $f(x) = 3x - 5$ dollars. ~~The company has 40 trucks and was founded in 1998.~~ What is $f^{-1}(7)$?

Target word: evaluate the inverse **Decides:** Solve $3x - 5 = 7$.

A	Dropped element	$(7 - 5)/3$ subtracts 5 instead of adding it back.
B	Correct	$3x - 5 = 7 \rightarrow 3x = 12 \rightarrow x = 4$.
C	Confusion set	Treats f^{-1} as the reciprocal of $f(7)$.
D	Inverted operation	That is $f(7)$, not the inverse.

3. Answer A Concept 2 · transfer

For a one-to-one function h , you know that $h^{-1}(10) = 4$. Which statement must be true?

Target word: reverse: from inverse value to function value **Decides:** $h^{-1}(b) = a$ means $h(a) = b$.

A	Correct	Inverses swap inputs and outputs.
B	Wrong stem number	Mixes the given numbers without the inverse relationship.
C	Confusion set	Treats the inverse as a reciprocal.
D	Inverted operation	Restates the inverse, not h .

4. Answer C Concept 3

A bacteria culture starts with 500 cells and doubles every 3 hours. ~~The lab keeps it at 37°C in a room with 6 incubators.~~ How many cells are there after 12 hours?

Target word: exponential growth **Decides:** $12 / 3 = 4$ doublings.

A	Adjacent category	Adds 500 each period (linear) instead of doubling.
B	Inverted operation	500×4 multiplies by the number of doublings instead of 2^4 .
C	Correct	$500 \times 2^4 = 8,000$.
D	Wrong stem number	500×2^{12} uses hours instead of doubling periods.

5. Answer A Concept 4

A Ferris wheel's position is described using angles on a unit circle. ~~The ride is 40 meters tall and carries 36 cars.~~ What is $\cos(2\pi/3)$?

Target word: cosine value in quadrant II **Decides:** Reference angle $\pi/3$; cosine negative in QII.

A	Correct	$\cos(\pi/3) = 1/2$; cosine is negative in quadrant II.
B	Dropped element	Uses the reference value but drops the negative sign.
C	Confusion set	That is $\sin(2\pi/3)$.
D	Adjacent category	Negative sine-style value; confuses sine and cosine.

6. Answer C Concept 5

A sound wave in a studio is modeled by $y = 3 \sin(4x) + 1$. ~~The studio has foam panels and was built last year.~~ What are the amplitude and period?

Target word: amplitude and period **Decides:** $A = 3$, $B = 4$.

A	Adjacent category	Adds the vertical shift of 1 to the amplitude.
B	Wrong stem number	Uses B as the amplitude and ignores it in the period.
C	Correct	$ A = 3$; $2\pi/4 = \pi/2$.
D	Inverted operation	Multiplies 2π by 4 instead of dividing.

7. Answer B Concept 5 • transfer

Which function has amplitude 5 and period π ?

Target word: reverse: build the function **Decides:** $B = 2\pi/\pi = 2$.

A	Wrong stem number	Period would be 4π .
B	Correct	Amplitude 5; period $2\pi/2 = \pi$.
C	Inverted operation	Period would be $2\pi/\pi = 2$.
D	Confusion set	Swaps amplitude and B.

8. Answer D Concept 6

A student club sells two items. Their records give $2x + 3y = 12$ and $x - y = 1$. ~~The club meets on Thursdays and has 15 members.~~ What is the solution (x, y) ?

Target word: solve the system **Decides:** Substitute $x = y + 1$.

A	Adjacent category	Satisfies only the first equation (the x-intercept).
B	Dropped element	Satisfies only $x - y = 1$; $2(1) + 3(0) = 2$, not 12.
C	Inverted operation	Swaps x and y ; fails $x - y = 1$.
D	Correct	$2(y + 1) + 3y = 12 \rightarrow 5y = 10 \rightarrow y = 2$, $x = 3$.

9. Answer D Concept 6 · transfer

For what value of k does the system $2x + y = 4$ and $4x + 2y = k$ have infinitely many solutions?

Target word: solve for the constant **Decides:** Second equation must be exactly twice the first.

A	Wrong stem number	Copies the first constant without scaling; the lines would be parallel.
B	Absolute language	Same slope with a different intercept gives no solution.
C	Inverted operation	Divides instead of multiplies.
D	Correct	$2 \times (2x + y = 4)$ gives $4x + 2y = 8$, the same line.

10. Answer D Concept 7

A park designer maps a circular fountain with $x^2 + y^2 - 6x + 4y - 12 = 0$, where units are meters. ~~The park opened in May and has 3 entrances.~~ What are the center and radius?

Target word: center and radius **Decides:** $(x - 3)^2 + (y + 2)^2 = 25$.

A	Dropped element	Forgets the square root.
B	Inverted operation	Flips the signs of the center.
C	Wrong stem number	Uses the constant 12 before completing the square.
D	Correct	$12 + 9 + 4 = 25$; $r = 5$.

11. Answer B Concept 8

A student saves \$5 in week 1 and increases her deposit by \$4 each week. ~~She keeps the money in a blue jar and works at a café on weekends.~~ How much does she deposit in week 20?

Target word: nth term of arithmetic sequence **Decides:** $a_{20} = 5 + 19 \times 4$.

A	Dropped element	20×4 drops the first deposit.
B	Correct	$5 + (20 - 1)(4) = 81$.
C	Confusion set	That is the total of 20 deposits, not the 20th deposit.
D	Wrong stem number	Uses $n = 20$ instead of $n - 1 = 19$.

12. Answer C Concept 8 · transfer

A geometric sequence starts 3, 6, 12, ... What is the sum of the first 6 terms?

Target word: geometric sum instead of a term **Decides:** $S_6 = 3(2^6 - 1)/(2 - 1)$.

A	Dropped element	3×2^6 drops the $- 1$ in the formula.
B	Confusion set	That is the 6th term, 3×2^5 .
C	Correct	$3 \times 63 = 189$.
D	Adjacent category	Treats it as arithmetic (3, 6, 9, 12, 15, 18), which sums to 63.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	7	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	1	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Function composition	$f(g(x))$ means evaluate g first, then put that output into f ; order matters.	Evaluating the outer function first, or multiplying f and g .	Nested notation such as $f(g(3))$.
2	Inverse functions	An inverse undoes the function: if $f(a) = b$ then $f^{-1}(b) = a$; find it by swapping x and y and solving.	Treating $f^{-1}(x)$ as the reciprocal $1/f(x)$.	f^{-1} notation or “undo.”
3	Exponential growth and logarithms	Repeated doubling multiplies by 2 each period: $A = A_0 \cdot 2^{(t/d)}$; a logarithm answers “what exponent gives this value?”	Adding a fixed amount each period (linear thinking).	Doubling time, half-life, or log notation.
4	Unit circle values and signs	Use the reference angle for the size and the quadrant for the sign: cosine is x , sine is y .	Using the reference-angle value without adjusting the sign.	An angle beyond 90° or $\pi/2$.
5	Amplitude and period of sinusoids	For $y = A \sin(Bx) + D$, amplitude = $ A $ and period = $2\pi/ B $; D shifts the midline only.	Multiplying 2π by B , or adding D to the amplitude.	A trig function with coefficients inside and outside.
6	Systems of linear equations	Solve by substitution or elimination; parallel lines (same slope, different intercept) have no solution, identical lines have infinitely many.	Stopping after finding one variable, or assuming every system has one solution.	Two equations in two unknowns.
7	Circle equations by completing the square	$(x - h)^2 + (y - k)^2 = r^2$ gives center (h, k) and radius r ; complete the square to reach this form.	Flipping the signs of the center or forgetting the square root for the radius.	x^2 and y^2 terms with equal coefficients plus linear terms.
8	Arithmetic vs. geometric sequences	Arithmetic: $a_n = a_1 + (n - 1)d$; geometric: $a_n = a_1 \cdot r^{(n - 1)}$, $S_n = a_1(r^n - 1)/(r - 1)$.	Using n instead of $n - 1$, or confusing a term with a sum.	A constant difference or a constant ratio.

Course scope (public description): Precalculus builds on previous math courses to provide a deeper understanding of fundamental mathematical concepts and problem-solving skills to prepare students for calculus and professional success. The course will lead students to engage with functions, trigonometry, systems of equations, analytic geometry, and sequences and series. Through interactive learning experiences and real-world applications, precalculus will help students develop a deeper understanding of mathematical principles and their practical significance across diverse fields. Successful completion of a college level algebra course is a prerequisite for this course.